The Method of Exhaustion

How Ancient Greek Geometry Bounded the Infinite

Pavel Shago

Saint Petersburg State University

History of Mathematics in English

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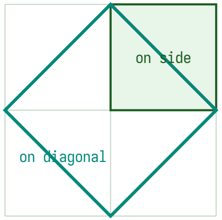
A Number That Was Not a Number

A Pythagorean drew the diagonal of a square and upon it raised a second square which covered twice the first.

Does a single measure then divide both side and diagonal, each a whole number of times? Any such measure forces a smaller, and smaller still, without end.



twice the square on the side

And this was the *easy* case. The circle posed a deeper problem. Its boundary is curved, its area is not polygonal, and its ratio of circumference to diameter would later prove to be no ratio of integers at all. How, then, could Greek geometry prove exact statements about circular area without treating infinity as an object?

The Barrier

Greek Geometry

Direct comparison
Cut a polygon, rearrange into another of known area

Similarity ratios
Areas of similar figures stand in the duplicate ratio of their sides

Both are finite. Neither works for a circle or a sphere.

Antiphon the Sophist (480–411 BCE), reported method

Antiphon inscribed regular polygons in a circle, repeatedly doubling the number of sides, arguing that the remaining gap would eventually be exhausted

So ”eventually equals” is an assertion about infinity and infinity is not an object of proof. **The real problem was proof-theoretic**, not numerical.

Eudoxus of Cnidus (390–337 BCE) answered with a new definition of equality of ratios of magnitudes

Ἐν τῷ αὐτῷ λόγῳ μεγέθη λέγεται εἶναι πρῶτον πρὸς δεύτερον καὶ τρίτον πρὸς τέταρτον, ὅταν τὰ τοῦ πρώτου καὶ τρίτου ἰσάκις πολλαπλάσια τῶν τοῦ δευτέρου καὶ τετάρτου ἰσάκις πολλαπλασίων καθ’ ὁποιονοῦν πολλαπλασιασμὸν ἑκάτερον ἑκατέρου ἢ ἅμα ὑπερέχῃ ἢ ἅμα ἴσα ᾖ ἢ ἅμα ἐλλείπῃ ληφθέντα κατάλληλα.

Or in modern notation: $a : b = c : d$ iff for all positive integers m, n

$$ma > nb \iff mc > nd, \qquad ma = nb \iff mc = nd$$

Two ratios are equal iff they agree on *every* rational test. The ratio is defined by its behaviour against all rationals no numerical value is ever produced.

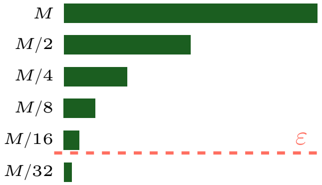
Modern echo. This is Dedekind’s construction of the real numbers (1872 CE). Eudoxus anticipated it by 2200 years.

One technical lemma does the heavy lifting. The *Archimedean halving property*, due to Eudoxus:

Elements, Book X, Proposition 1

If from any magnitude there be subtracted a part not less than its half, and from the remainder a part not less than its half, and so on continually, there will at length remain a magnitude less than any assigned magnitude of the same kind

Modern restatement: for every $\varepsilon > 0$ and every magnitude M, there exists n with $\left(\frac{1}{2}\right)^n M < \varepsilon$. Finite. No "as $n \to \infty$." Rather: **for every tolerance, eventually.**



Modern echo. The ε-δ discipline of Cauchy (1821 CE) and Weierstrass, 2300 years early.

The Exhaustion Schema

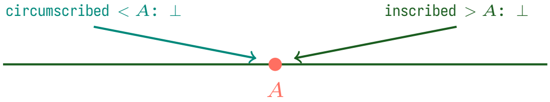
To prove the area of a curved figure equals A rule out both strict inequalities.

If area $> A$

Inscribe polygons, halving the uncovered region each step, until the remainder falls below A. The inscribed polygon then exceeds A which is impossible

If area $< A$

Circumscribe polygons, their excess shrinks below A. The circumscribed polygon then falls below A. Same contradiction, reversed



Therefore the area equals A. **No limit is ever taken.**
Modern echo. Riemann integration: $\underline{S}_n \leq \int f \leq \overline{S}_n$, and the gap shrinks.

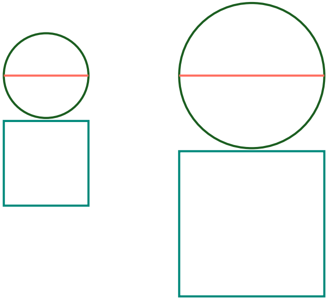
Elements, Book XII, Proposition 2

Circles are to one another as the squares on their diameters

Notice what this result *is* and *is not*

- **Not** a value of π. No number for circular area ever appears.
- **A structural law:** circles behave *relative to each other*, scaling as squares on any linear dimension.

The constant that relates a circle to a square is a separate, harder question left to Archimedes.



circles :: squares on diameters

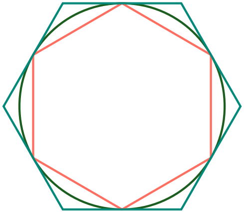
***Measurement of a Circle*, Proposition 3 (250 BCE)**

The perimeter of every circle exceeds three times the diameter by less than one-seventh, and by more than ten seventy-oneths, of the diameter

Archimedes applied exhaustion to a 96-gon (six doublings from a hexagon) bounding the ratio above and below.

Modern restatement: $3 + \frac{10}{71} < \pi < 3 + \frac{1}{7}$

Four-digit accuracy with a *proven enclosure* it is not even a measurement, a theorem about the ratio location.



inscribed (lower bound)
circumscribed (upper bound)

What Archimedes Did Not Publish

Archimedes wrote a private letter to Eratosthenes —*The Method of Mechanical Theorems* —lost for over 1000 years, recovered in 1906 from a Byzantine palimpsest.

In it, he sliced figures into *indivisibles* (infinitely many parallel lines) and *balanced them on a lever* to guess areas and volumes. Mechanical intuition, not geometric proof.

> “It is easier to find the proof once one has acquired some knowledge of the questions by the method.” —Archimedes to Eratosthenes

Archimedes used indivisibles to **discover**, then rewrote results as exhaustion proofs to **publish**. Discovery and certification separated —the same split between numerical experiment and formal verification today.

Quadrature of the Parabola: the area of a parabolic segment equals $\frac{4}{3}$ of its inscribed triangle T.

Modern view:

$$T\left(1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}+\cdots\right)=\frac{4}{3}T.$$

Archimedes' view: for every n, he filled the segment with triangles of total area

$$T\left(1+\frac{1}{4}+\cdots+\frac{1}{4^n}\right) \quad \text{with remainder at most} \quad \frac{1}{3}\cdot\frac{1}{4^n}T.$$

He never wrote an infinite sum. He wrote a *finite identity* plus an *error bound that can be driven below any magnitude*. Same answer; no appeal to infinity; bulletproof Greek proof.

Modern echo. Partial sum plus remainder bound is still how convergence is proved.

The Meta-Mathematical Mechanism

Three commitments of Greek mathematics made the method possible:

1. **Magnitude over number.** Ratios of geometric magnitudes, not decimal values, were the primary objects. Irrationals were not a problem to be solved but a type to be handled.
2. **Proof over computation.** An approximation without a certified error bound was worthless. A bound without a computed value was a theorem.
3. **Finite control over infinite process.** A shrinking sequence is handled by the law it obeys, never by the limit point it approaches.

The method of exhaustion is these three commitments made procedural. Remove any one and the method collapses: it is inseparable from the philosophy of proof that produced it.

Modern Descendants

The schema is the operational core of much of modern mathematics:

- **Real numbers** (Dedekind, 1872) —direct formalisation of Eudoxus' ratios.
- ε-δ **analysis** (Cauchy, Weierstrass) —the Archimedean lemma as the definition of limit.
- **Riemann integration** —the curved region bracketed by lower and upper sums; the gap shrinks.
- **Interval arithmetic & verified numerics** —Archimedes' two-sided enclosure, now standard in computer-assisted proofs.
- **A posteriori error estimates in FEM** —approximate the solution, certify a bound. Industrial scale.

In every case the ancient pattern is preserved: approximate by simpler objects, and *prove the error shrinks*.

Conclusion

The method of exhaustion is often called "ancient integral calculus." That label is too loose and too modest.

> The Greek breakthrough was not a calculus algorithm. It was a **proof discipline for infinite processes that never admits infinity as an object.**

That discipline —approximate, bound, contradict —is still the proof discipline of modern numerical analysis and real analysis. Greek mathematics did not find a way to *compute* curves; it found a way to *reason* about them. The computing came later; the reasoning is still ours.

Thanks for your attention